

Energybae

1. What I Built

I built a fully browser-based Solar Load Calculator web app that automates the entire electricity bill-to-Excel workflow. A user uploads their electricity bill (JPG, PNG, or PDF), the app reads the bill using Google Gemini Vision AI, automatically fills in all key fields (consumer name, units consumed, bill amount, meter number, etc.), calculates the recommended solar system size, savings, payback period, and environmental impact — and finally lets the user download a ready-to-use Excel report with 3 sheets (Bill Data, Solar Calculation, Summary).

2. Tools & Technologies Used

Tool / Tech	Purpose
HTML + CSS + JavaScript	Complete frontend UI — no framework needed, runs in any browser
Google Gemini 1.5 Flash API	Vision AI to read and extract data from bill images/PDFs
SheetJS (xlsx.js)	Generate and download the filled Excel file client-side
Google Fonts (DM Sans)	Clean, professional typography for the UI

3. How It Works (4 Steps)

- Step 1 — Upload Bill:** User drags and drops or selects an electricity bill (JPG, PNG, or PDF).
- Step 2 — AI Extraction:** Gemini 1.5 Flash reads the bill image and returns structured JSON with all key fields — consumer name, units consumed, bill amount, meter number, tariff, readings, etc.
- Step 3 — Review & Calculate:** Extracted data is auto-filled into an editable form. Solar calculations run instantly: system size (kWp), number of panels, rooftop area, monthly savings, payback period, 25-year projection, and CO2 offset.
- Step 4 — Download Excel:** One click generates a professional 3-sheet Excel file — Bill Data, Solar Calculation (with 25-year table), and a one-page Summary — ready to share with the customer.

4. What I Would Improve Next

- Add support for multiple bills at once (batch processing) so the team can handle many customerstogether.
- Connect directly to Google Sheets so the output auto-saves to the team's shared drive withoutmanual downloading.

- Train a custom model specifically on MSEDCL and other Indian utility bill formats for even higher extraction accuracy.
- Add a customer management dashboard to track all submissions, solar recommendations, and follow-ups in one place.

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